PowerBi Dax Basics

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Agenda

DAX Fundamentals

- Calculated Columns
- Calculated Measures

DAX Evaluation Context

- Row Context
- Filter Context

Time Series Analysis

- YTD Sales
- Prior Year

Semi-Additive Measures

- Closing Balance
- Opening Balance

Context Transition

- Adding Row Filters to Filter Context

DAX

What is DAX?

- Data Analysis Expressions
- Expression language for Power BI, Power Pivot and SSAS Tabular

Why DAX?

- Designed to Support a Larger User Base

Introduction to Power BI DAX

What is DAX?

DAX (Data Analysis Expressions) is a formula language used in **Power BI**, **Excel Power Pivot**, and other Microsoft analytics tools to create custom calculations and analytical logic.

- Calculated Columns
- Measures
- Calculated Tables

Purpose of DAX

To analyze data and create dynamic, context-aware calculations that support business insights and decision-making in Power BI.

Why DAX?

- Designed to Support a Larger User Base
- Simpler than Traditional Technical Languages to Learn
- Leverage Existing Knowledge of Excel Formulas
- Less of a Learning Curve for Analyst

Why Learn DAX?

- DAX is essential for advanced data calculations and custom business metrics in Power BI.
- It enables time-based analysis such as year-to-date (YTD), quarter-over-quarter, and growth calculations.
- DAX allows analysts to define meaningful measures, including profit margin, growth rate, and performance indicators.
- It transforms raw data into actionable insights that support data-driven decision making.

Types of DAX Functions

DAX provides a wide range of built-in functions that support business analytics. These functions are commonly grouped by their analytical purpose.

- Aggregation functions
- Date and time functions
- Logical functions
- Filter and context functions

Teaching Note

In practice, DAX measures combine multiple function types within a single expression.

Common Types of DAX Functions

Function Type	Purpose	Examples
Aggregation	Summarize data by calculating totals, averages, counts, or minimum and maximum values.	SUM, AVERAGE, COUNT
Date and Time	Extract or manipulate date values for time-based analysis.	YEAR, MONTH, TODAY
Logical	Apply conditional logic and decision rules in calculations.	IF, SWITCH
Filter and Context	Modify evaluation context and control how filters affect calculations.	CALCULATE, ALL

Key Insight

Business measures often rely on **CALCULATE** to combine aggregation, filtering, and time logic.

New Columns

Full Name = ’Customer’[First Name] & ” ” & ’Customer’[Last Name]

Age Restrictions Logic

```
Age Restrictions = IF ( 'Customer'[Age] >= 55, "Old", IF ( 'Customer'[Age] >= 44,
"Middle age", IF ( 'Customer'[Age] >= 25, "Young", IF ( 'Customer'[Age] >= 15,
"Youngster"
) ) ) )

Month year = FORMAT('Date'[Date], "MM-YYYY")
```

```
Region Volume =
SWITCH(TRUE(),
    'Sales Territory'[Total Transactions] >= 7000, "High Volume",
    'Sales Territory'[Total Transactions] >= 4000, "Medium Volume",
    'Sales Territory'[Total Transactions] >= 1, "Low Volume",
    "NA" )
```

Figure: Enter Caption

PART-2

PART-3

PART-5

ASSIGNMENT